INFORMATION TECHNOLOGY AND INFRASTRUCTURE PROJECT PORTFOLIO

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OVERVIEW

This portfolio showcases my hands-on experience in IT support, networking, and cybersecurity through academic and self-driven projects. These projects demonstrate my ability to configure systems, design networks, troubleshoot technical issues, and analyze security vulnerabilities. Each project includes documentation, methodology, and outcomes to reflect real-world IT practices.

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1. Technical Skills

Systems & Networking:

- Active Directory, Windows Server, DHCP, DNS, Hyper-V
- TCP/IP, VLANs, Routing & Switching

Tools & Platforms:

- Cisco Packet Tracer
- Wireshark
- PowerShell
- Intune / Ivanti

Security:

- Vulnerability Analysis
- Access Control
- Endpoint Security

Project 1: Active Directory Domain Controller

Project Overview

Configured a Windows Server-based Active Directory Domain Controller within a virtual lab environment to establish centralized user administration, authentication, and network services. Implemented directory services, DNS, DHCP, and Group Policy to simulate an organizational IT environment.

Objectives

- Establish and configure Active Directory Domain Services (AD DS).
- Create and manage users, security groups, and organizational units.
- Configure DNS and DHCP services.
- Implement Group Policy Objects (GPOs) to manage user and system configurations.
- Configure Windows Firewall and access control policies.

Technologies Used

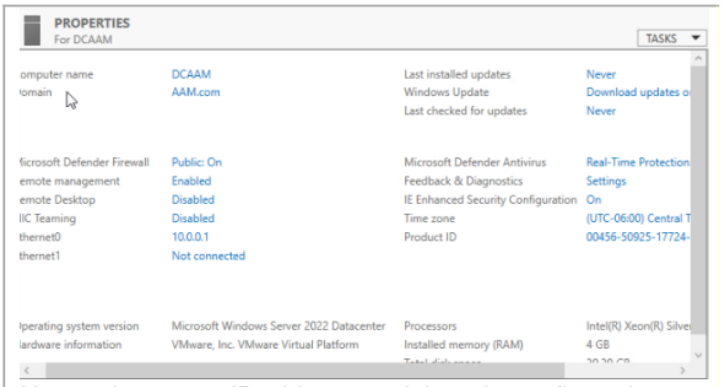
- Windows Server
- Active Directory Domain Services (AD DS)
- Hyper-V
- DNS and DHCP
- PowerShell
- Windows Firewall

Key Tasks Performed

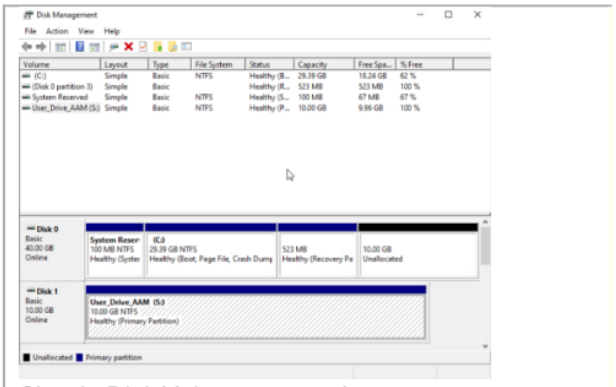
- Installed and configured Active Directory Domain Services.
- Created user accounts, security groups, and organizational units.
- Configured and applied Group Policy Objects to manage user permissions and system restrictions.
- Established DNS forward lookup zones and configured A records.
- Created DHCP scopes and IP address reservations.
- Configured folder redirection and password policies.
- Implemented Windows Firewall rules to control inbound and outbound network traffic.
- Used PowerShell to configure virtual networking components and manage virtual machine access.

Outcome

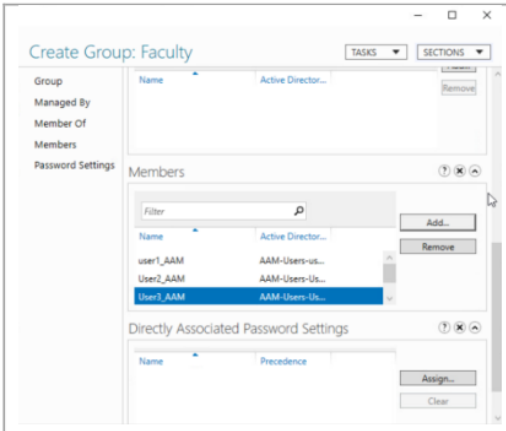
Successfully established a Windows Server-based domain environment demonstrating centralized identity management, user administration, network services, and policy enforcement. Gained practical experience configuring and managing core Active Directory infrastructure and Windows Server administrative functions.



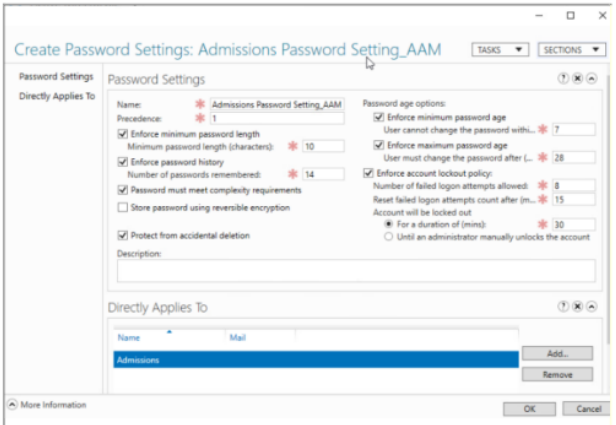
Name, time zone, IP address, and domain configuration for the domain controller.



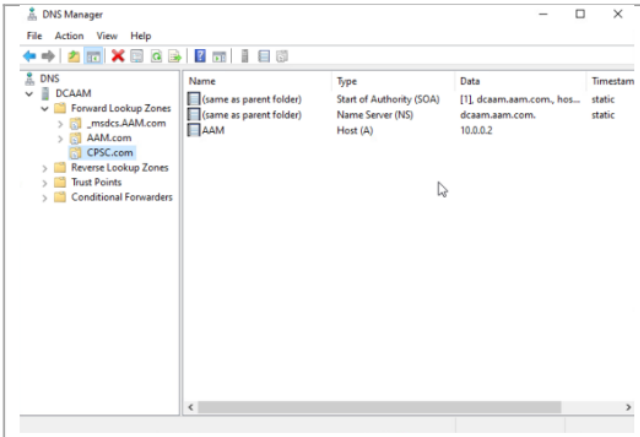
Simple Disk Volume created.



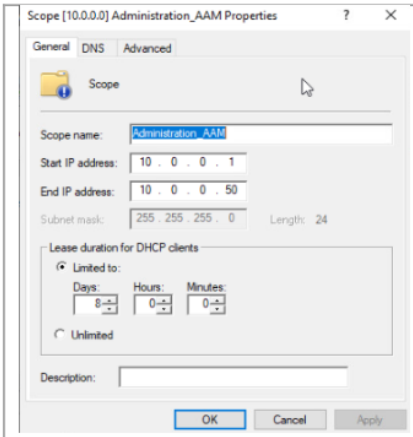
New users created; New group named faculty created; All three users added to the faculty group.



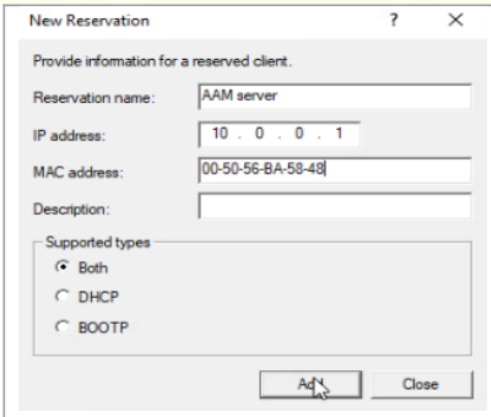
Group named Admissions also created; Password policy created and applied to the admission group.



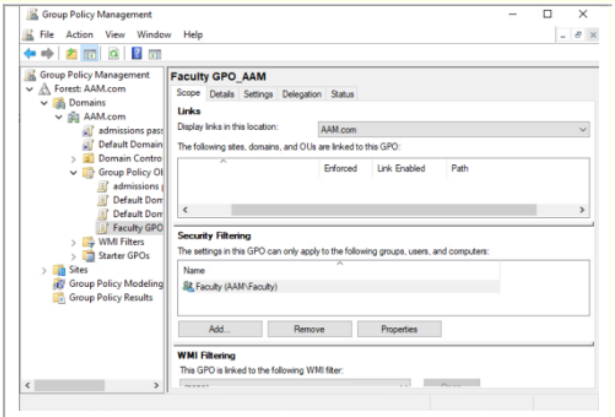
CPSC.com primary forward lookup zone was created; An A record in the CPSC.com zone was created.



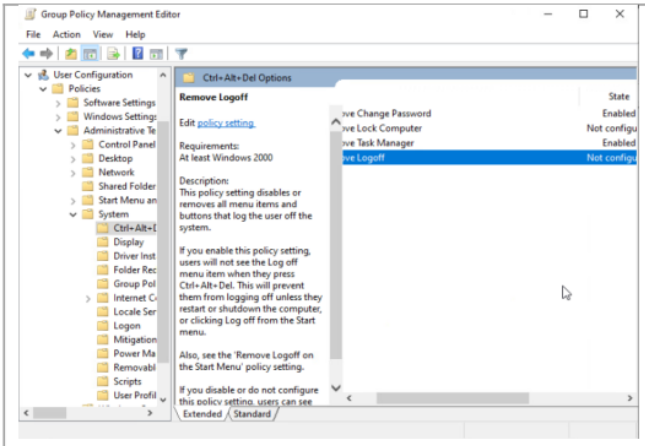
New IPv4 Scope created for address assignment to machines.



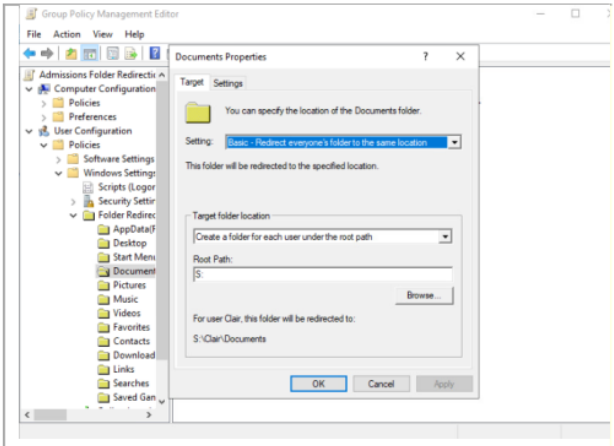
IPv4 Reservation created for the server.



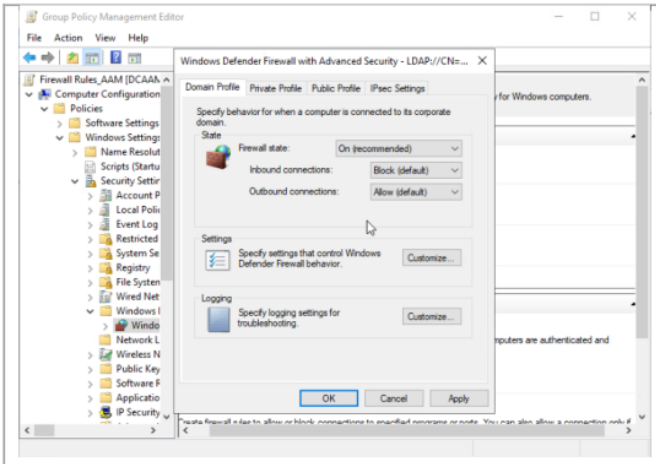
New GPO named Faculty GPO_AAM applied to the faculty group.



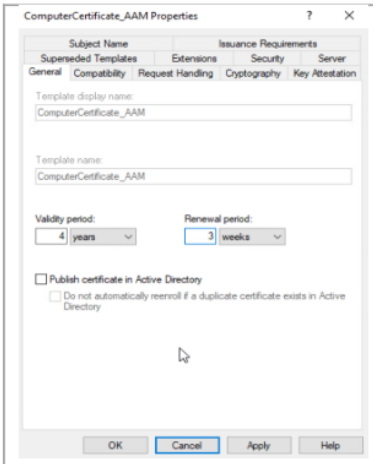
Remove task manager and remove change password are enabled as a GPO.



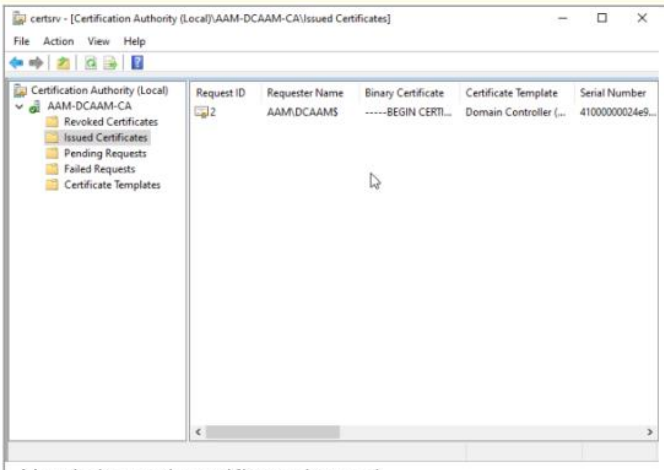
Folder redirection created for the admissions group.



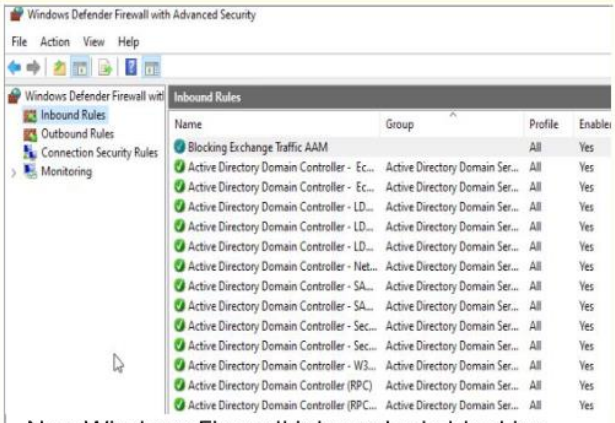
New GPO Firewall Rules AAM that blocks inbound and allows outbound connections.



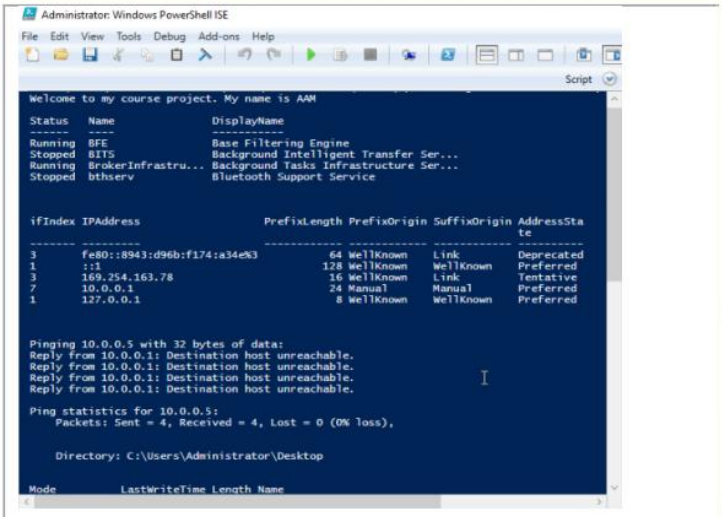
New certificate created.



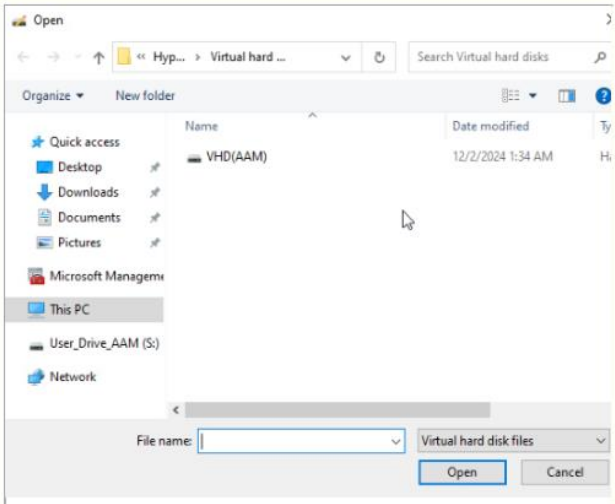
Newly issued certificate issued.



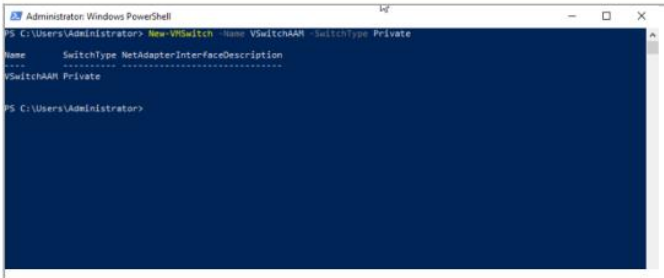
New Windows Firewall inbound rule blocking port 25.



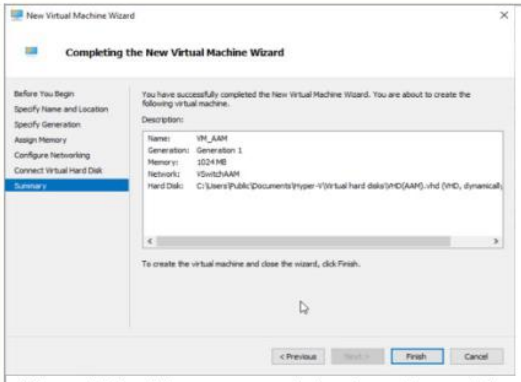
Results of a PowerShell script made to display certain text, and return certain information such as running programs and ping information.



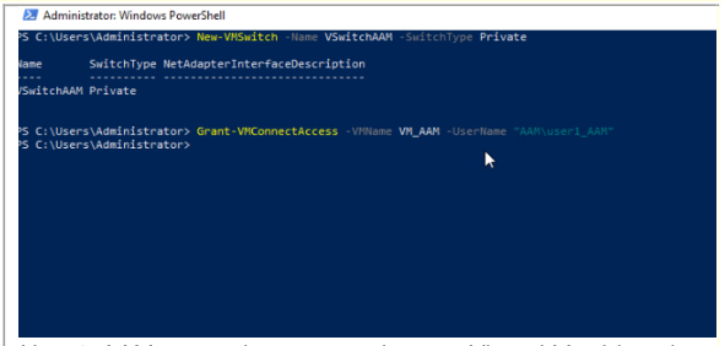
New Virtual Hard Disk created for memory allocation to the virtual machine.



New Virtual Domain switch created via Powershell.



New Virtual Machine created via virtual machine wizard.



User1_AAM granted access to the new Virtual Machine via PowerShell.

Project 2: Network Design and Configuration (Cisco Packet Tracer)

Project Overview

Designed and configured simulated network environments using Cisco Packet Tracer to develop practical networking skills. Worked with routers, switches, and end devices to establish connectivity, configure network interfaces, implement VLANs, and practice IPv4 and IPv6 addressing.

Objectives

- Design and document logical and physical network topologies.
- Configure routers, switches, and endpoint devices.
- Implement IP addressing and subnetting.
- Configure VLANs and switch interfaces.
- Establish and verify connectivity between network devices.
- Develop an understanding of physical network infrastructure and cabling standards.

Technologies Used

- Cisco Packet Tracer
- Cisco IOS CLI
- IPv4 and IPv6
- VLANs
- SLAAC
- Ethernet and Serial Interfaces

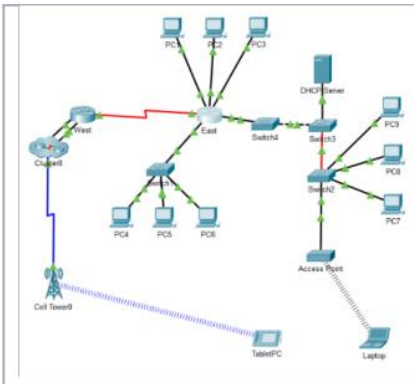
Key Tasks Performed

- Designed network topologies incorporating routers, switches, and endpoint devices.
- Identified physical connections, interface assignments, and appropriate cable types.
- Configured switch management interfaces and VLAN 1 addressing.
- Implemented basic switch configurations, including console access, passwords, and banner messages.
- Assigned IPv4 addresses and subnet masks to network devices.
- Configured router interfaces and examined interface status using Cisco IOS commands.
- Subnetted an IPv6 network and assigned network addresses to router interfaces.
- Configured automatic IPv6 address assignment using Stateless Address Autoconfiguration (SLAAC).
- Verified connectivity between endpoint devices using ICMP ping tests.

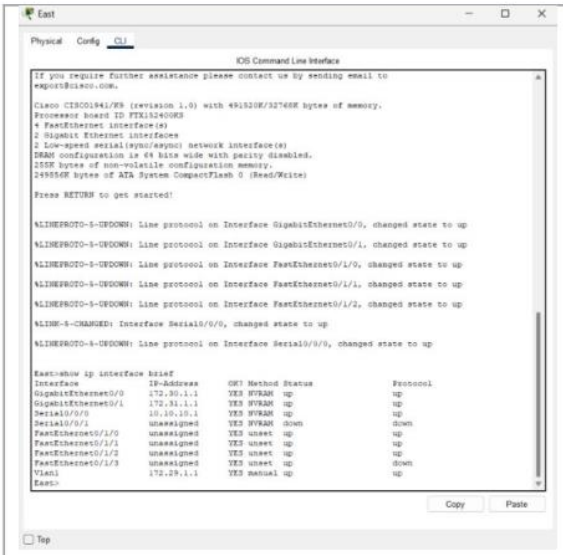
Outcome

Successfully designed and implemented simulated LAN environments demonstrating foundational networking concepts, device configuration, IP addressing, subnetting, and network connectivity. Developed practical familiarity with Cisco IOS commands, network topology design, and IPv4/IPv6 infrastructure.

a. Physical Layer and Cabling



The logical topology showing all physical connections.

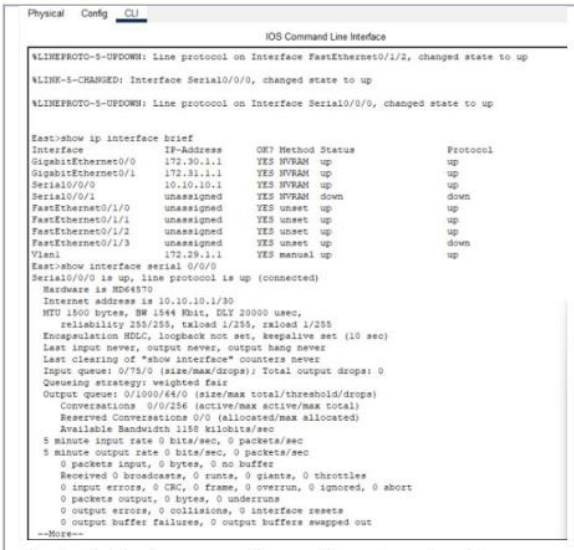


Ip interface brief command Configuration of all interfaces on the east router.

PT Activity: 00:46:20

Device	Interface	Cable Type	Device	Interface
East	GigabitEthernet0/0	Copper Straight-Through	Switch1	GigabitEthernet0/1
East	GigabitEthernet0/1	Copper Straight-Through	Switch4	GigabitEthernet0/1
East	FastEthernet0/1/0	Copper Straight-Through	PC1	FastEthernet0
East	FastEthernet0/1/1	Copper Straight-Through	PC2	FastEthernet0
East	FastEthernet0/1/2	Copper Straight-Through	PC3	FastEthernet0
Switch1	FastEthernet0/1	Copper Straight-Through	PC4	FastEthernet0
Switch1	FastEthernet0/2	Copper Straight-Through	PC5	FastEthernet0
Switch1	FastEthernet0/3	Copper Straight-Through	PC6	FastEthernet0
Switch4	GigabitEthernet0/2	Copper Cross-Over	Switch3	GigabitEthernet3/1
Switch3	GigabitEthernet5/1	Fiber	Switch2	GigabitEthernet5/1
Switch2	FastEthernet0/1	Copper Straight-Through	PC7	FastEthernet0
Switch2	FastEthernet1/1	Copper Straight-Through	PC8	FastEthernet0
Switch2	FastEthernet2/1	Copper Straight-Through	PC9	FastEthernet0
Switch2	Gigabit3/1	Copper Straight-Through	AccessPoint	Port 0
East	Serial0/0/0	Serial DCE (connect to East first)	West	Serial0/0/0

Each physical connections cable type and the interface it's connected to.



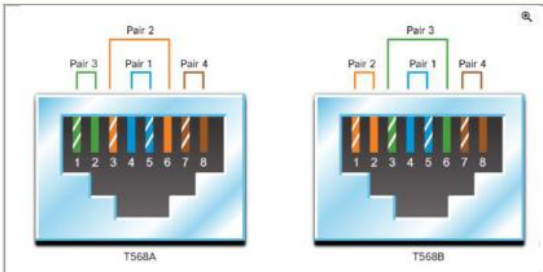
Serial interface configuration showing the bandwidth: Bandwidth on serial interfaces is used by routing processes to determine the best path to a destination. It does not indicate the actual bandwidth of the interface. Actual bandwidth is negotiated with a service provider.



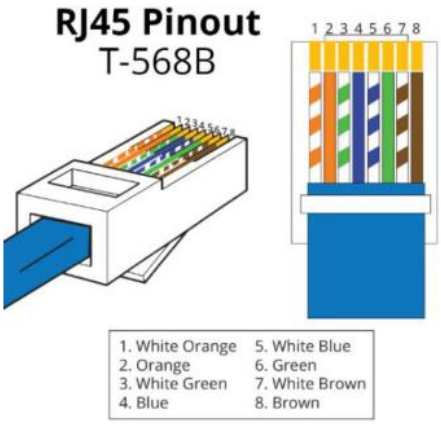
Physical view of switch 2 and it's interfaces.



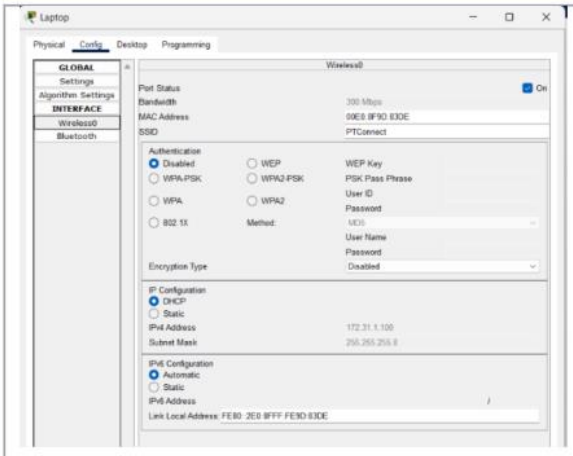
Physical description of modules that can be found on cisco switches and routers.



Physical view of T568A and T568B twisted pair wiring.

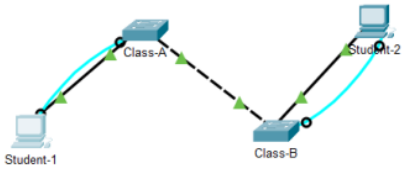


Physical view of T568B unshielded twisted pair wiring terminated by an RJ45 connector.



Configured wireless connection on Laptop.

b. Basic Switch and Endpoint Configuration



Logical topology of a simple LAN

Device	Interface	IP Address	Subnet Mask
Class-A	VLAN 1	172.16.5.35	255.255.255.0
Class-B	VLAN 1	172.16.5.40	255.255.255.0
Student-1	NIC	172.16.5.50	255.255.255.0
Student-2	NIC	172.16.5.60	255.255.255.0

Objectives

- Configure hostnames and IP addresses on two Cisco Internetwork Operating System (IOS) switches using the command-line interface (CLI).
- Use Cisco IOS commands to specify or limit access to the device configurations.
- Use IOS commands to save the running configuration.
- Configure two host devices with IP addresses.
- Verify connectivity between the two PC end devices.

Scenario

As a recently hired LAN technician, your network manager has asked you to demonstrate your ability to configure a small LAN. Your tasks include configuring initial settings on two switches using the Cisco IOS and configuring IP address parameters on host devices to provide end-to-end connectivity. You are to use two switches and two hosts/PCs on a cabled and powered network.

Instructions

Configure the devices to fulfill the requirements below.

Requirements

- Use a console connection to access each switch.
- Name **Class-A** and **Class-B** switches.
- Use the **R4Xe3** password for all lines.
- Use the **C4aJa** secret password.
- Encrypt all clear text passwords.
- Configure an appropriate message-of-the-day (MOTD) banner.
- Configure addressing for all devices according to the Addressing Table.
- Save your configurations.
- Verify connectivity between all devices.

Configuration objectives and requirements for the LAN.

```
lass-A#configure terminal
nter configuration commands, one per line.  End with CNTL/Z.
lass-A(config)#interface vlan1
lass-A(config-if)#ip address 172.16.5.35 255.255.255.0
lass-A(config-if)#no shutdown
lass-A(config-if)#exit
lass-A(config)#exit
lass-A#
SYS-5-CONFIG_I: Configured from console by console

lass-A#copy running-config startup-config
estination filename [startup-config]?
uilding configuration...
OK]
lass-A#
```

Class A switch VLAN1 configuration.
Class B switch is configured similarly with its own respective IP address.

```
Press RETURN to get started!

Authorized access only !

User Access Verification

Password:
Password:

Class-A>enable
Password:
Class-A#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Class-A(config)#show startup-config
                ^
% Invalid input detected at '^' marker.

Class-A(config)#exit
Class-A#
%SYS-5-CONFIG_I: Configured from console by console

Class-A#show startup-config
Using 1298 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Class-A
!
!
enable secret 5 $1$mERr$3VSsmT5sO0ky2/3RzW2XS1
enable password 7 081318760C4A
```

Start up configuration showing the configuration of the banner message of the day, the line console password, secret password, and password encryption.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.5.60

Pinging 172.16.5.60 with 32 bytes of data:

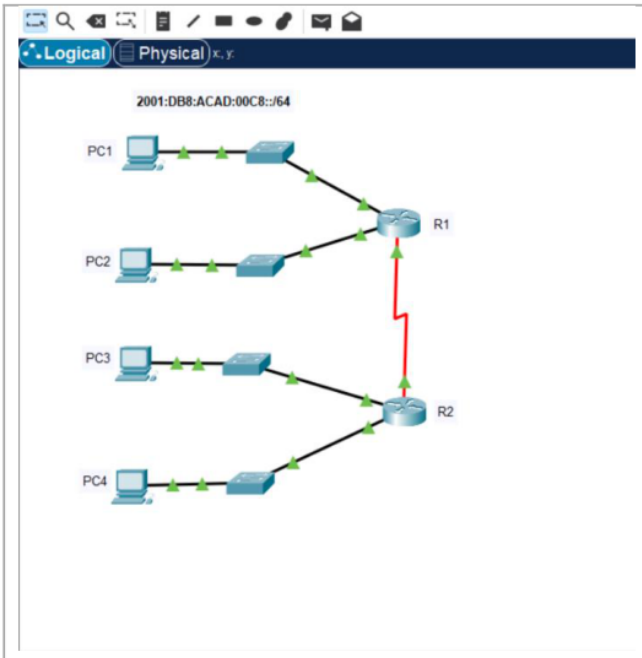
Reply from 172.16.5.60: bytes=32 time=2ms TTL=128
Reply from 172.16.5.60: bytes=32 time<1ms TTL=128
Reply from 172.16.5.60: bytes=32 time<1ms TTL=128
Reply from 172.16.5.60: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.5.60:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

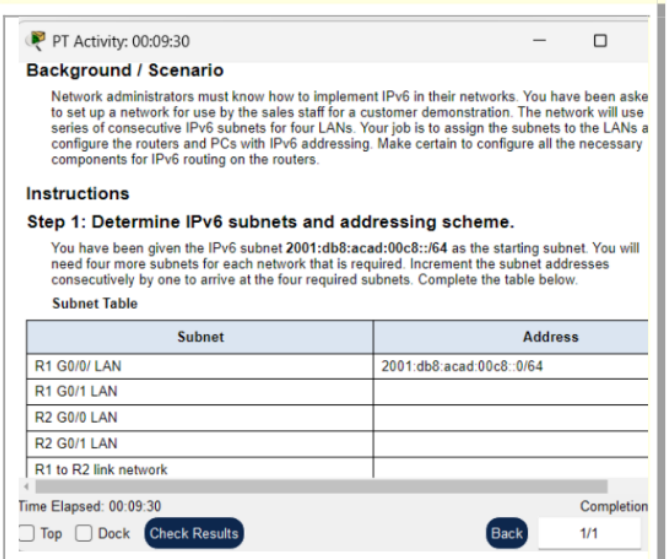
C:\>
```

Ping results from Student 1 PC to Student 2 PC to verify connectivity.

c. Subnetting an IPv6 LAN



Logical Topology of a LAN running IPv6.



The objective and the Network address that needs to subnetted as well as the number of subnets needed.

IOS Command Line Interface

```
R1>clear history
          ^
% Invalid input detected at '^' marker.

R1>
R1>enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface gigabitethernet 0/0
R1(config-if)#ipv6 address 2001:db8:acad:00c8::1/64
R1(config-if)#ipv6 address fe80::1 link-local
R1(config-if)#enable ipv6
          ^
% Invalid input detected at '^' marker.

R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface gigabitethernet 0/1
R1(config-if)#ipv6 address 2001:db8:acad:00c9::1/64
R1(config-if)#ipv6 address fe80::1 link-local
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface serial 0/0/0
R1(config-if)#ipv6 address 2001:db8:acad:00cc::1/64
R1(config-if)#ipv6 address fe80::1 link-local
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Ip address assignment of the required interfaces on Router #1. Two of the subnetted network addresses have been assigned to R1's interfaces.

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

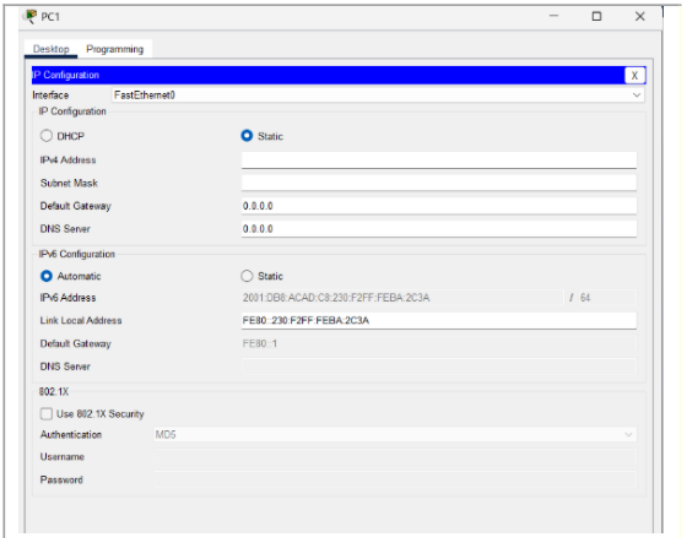
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up


R2>enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface gigabitethernet 0/1
R2(config-if)#ipv6 address 2001:db8:acad:00ca::1/64
%GigabitEthernet0/1: Error: 2001:DB8:ACAD:CA::/64 is overlapping with 2001:DB8:ACAD:CA::/
GigabitEthernet0/0
R2(config-if)#exit
R2(config)#interface gigabitethernet 0/0
R2(config-if)#ipv6 address 2001:db8:acad:00ca::1/64
R2(config-if)#ipv6 address fe80::2 link-local
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface gigabiethernet 0/1
                  ^
% Invalid input detected at '^' marker.

R2(config)#interface gigabitethernet 0/1
R2(config-if)#ipv6 address 2001:db8:acad:00cb::1/64
R2(config-if)#ipv6 address fe80::2
% Incomplete command.
R2(config-if)#ipv6 address fe80::02 link-local
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface serial 0/0/0
R2(config-if)#ipv6 address 2001:db8:acad:00cc::2/64
R2(config-if)#ipv6 address fe80::2 link-local
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
```

Ip address assignment of the required interfaces on Router #2. Two of the subnetted network addresses have been assigned to R2's interfaces.



PC1's automatic IP address assignment using SLAAC.

Project 3: Red and Blue Team Security Exercises

Project Overview

Conducted hands-on cybersecurity exercises within a controlled lab environment to examine network vulnerabilities, demonstrate common attack techniques, and implement security countermeasures. Focused on securing web communications, configuring remote access VPNs, strengthening SSH authentication, and implementing encrypted communication using digital signatures.

Objectives

- Identify and analyze network and system vulnerabilities.
- Understand common attack techniques and their potential impact.
- Implement encryption and secure communication protocols.
- Configure secure remote access and authentication mechanisms.
- Apply security controls to strengthen system integrity and access protection.

Technologies Used

- Linux
- Ettercap
- Nginx
- SSH
- Windows Server Routing and Remote Access (RRAS)
- OpenVPN
- pfSense
- Kleopatra
- Digital Certificates and Public/Private Key Cryptography

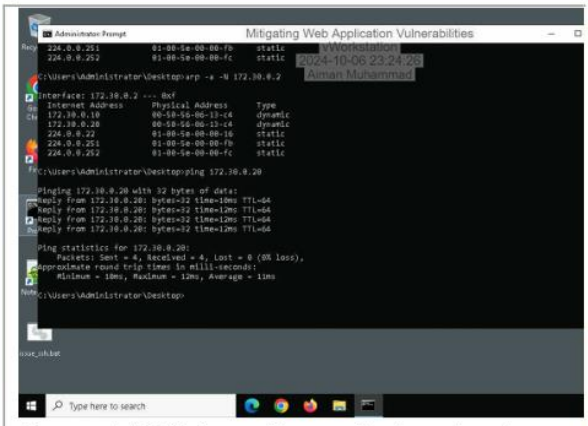
Key Tasks Performed

- HTTP MITM analysis: Used Ettercap to capture and examine unencrypted HTTP traffic in a controlled environment, demonstrating the exposure of sensitive information. Subsequently configured HTTPS using Nginx and implemented certificate-based security.
- SSH hardening: Configured SSH for secure remote administration and modified authentication settings to restrict password-based access, requiring key-based authentication.
- Remote access VPN configuration: Implemented Windows Server RRAS and OpenVPN to establish remote connectivity. Configured VPN address allocation, access policies, and firewall settings using Windows Defender Firewall and pfSense.
- Log analysis: Examined Linux and Windows Server log entries to observe and verify system and connection activity.
- Digital signatures and encryption: Generated public and private key pairs, created digital signatures, and used Kleopatra to demonstrate certificate creation, message encryption, and successful decryption.

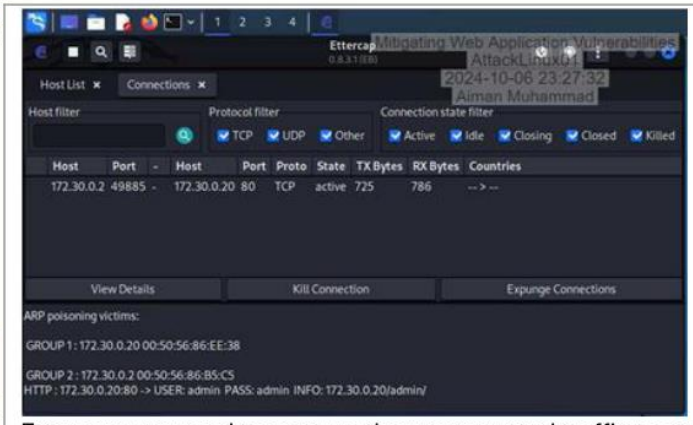
Outcome

Successfully demonstrated several cybersecurity concepts through practical implementation and testing. Gained hands-on experience analyzing unencrypted traffic, strengthening SSH authentication, configuring remote access VPNs, managing digital certificates, and applying encryption techniques. Developed a stronger understanding of network security, authentication, and defensive security practices.

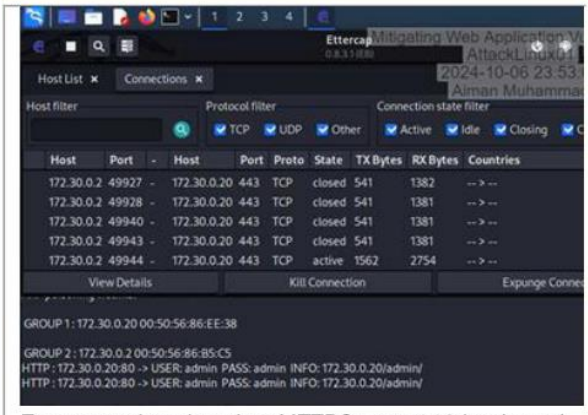
a. Mitigate an HTTP MITM Attack by Using HTTPS



Ping and ARP information on the target web server running HTTP from a Linux machine.



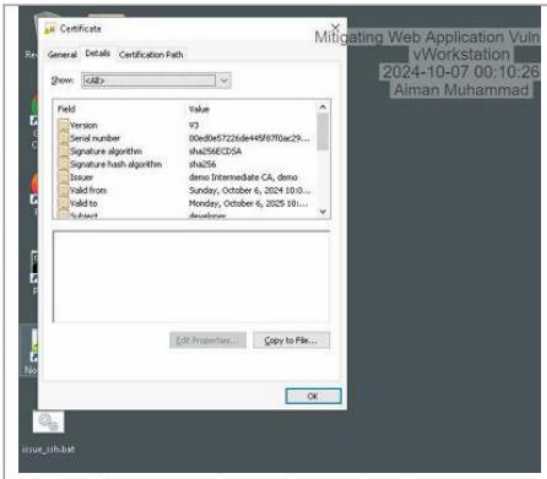
Ettercap was used to capture the unencrypted traffic to and from the target web server running HTTP on a Linux machine. This screen capture shows the username and password.



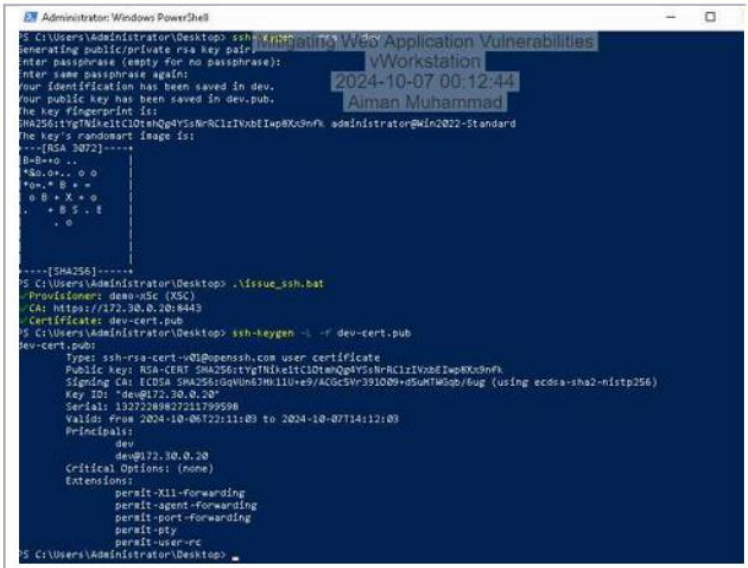
Ettercap showing that HTTPS port 443 is closed for this web server.



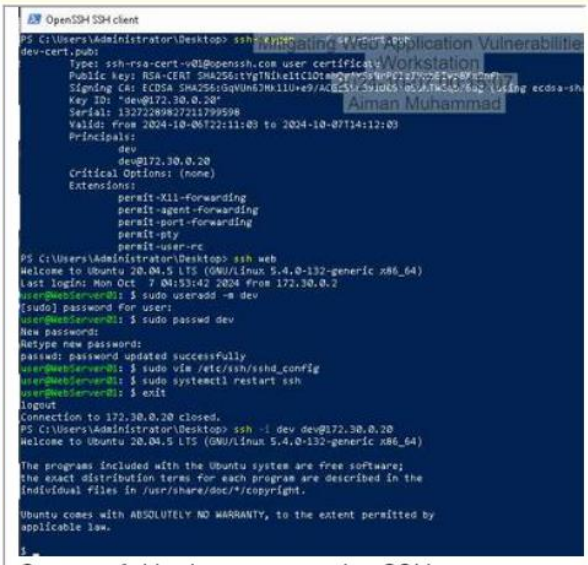
Script to update the web server to HTTPS and install the new nginx version.



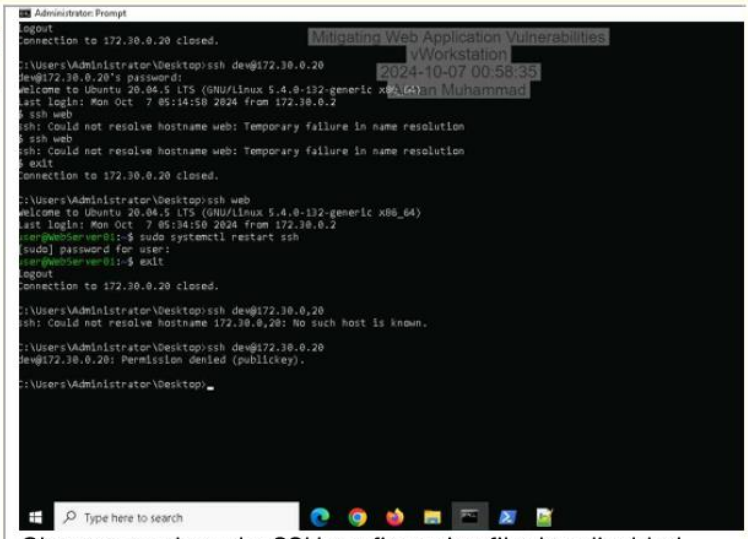
A certificate was issued to to harden access control.



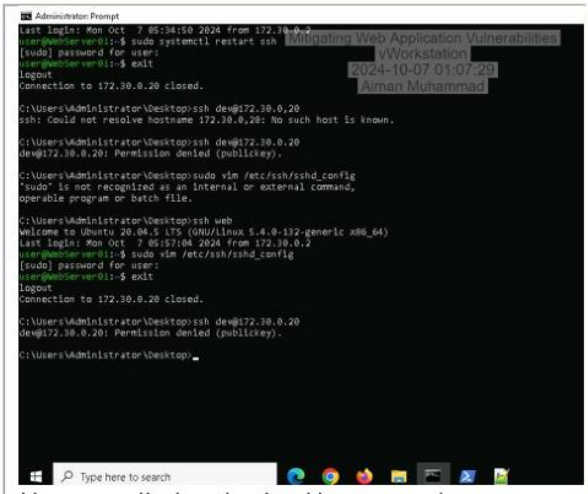
SSH configured on the web server for secure remote access.



Successful login attempt using SSH.

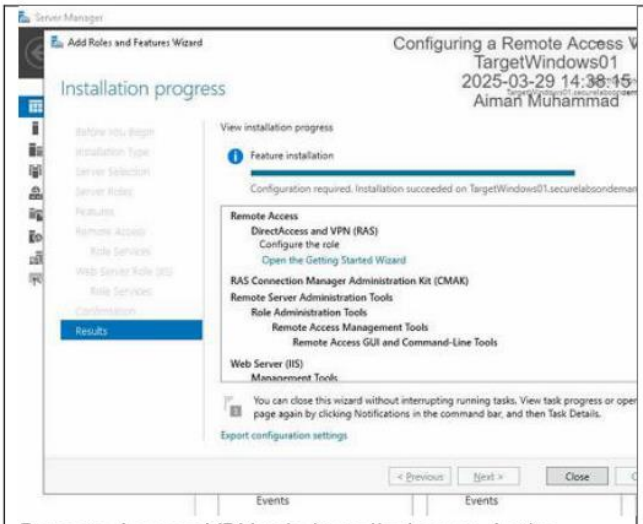


Changes made to the SSH configuration file that disabled password access for user Dev was disabled. Access is only allowed using a certificate or key. This screen capture shows an unsuccessful login attempt when no key or certificate is used.

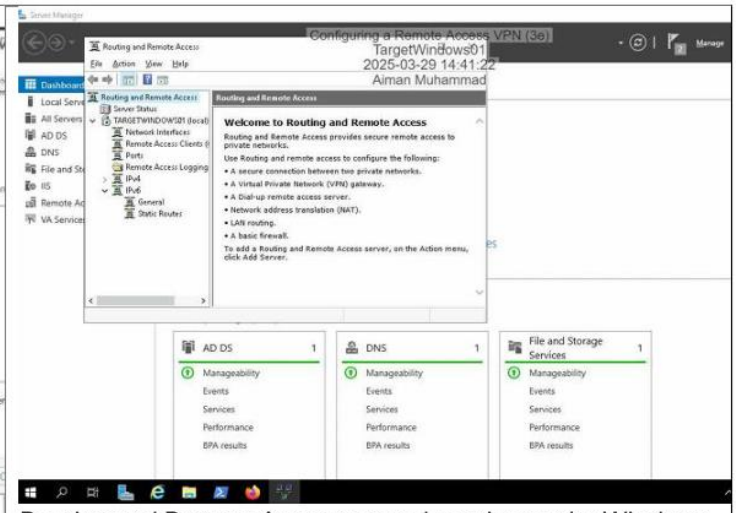


User supplied authorized key were also disabled in the configuration file. This screen capture shows an unsuccessful login attempt when no certificate is used.

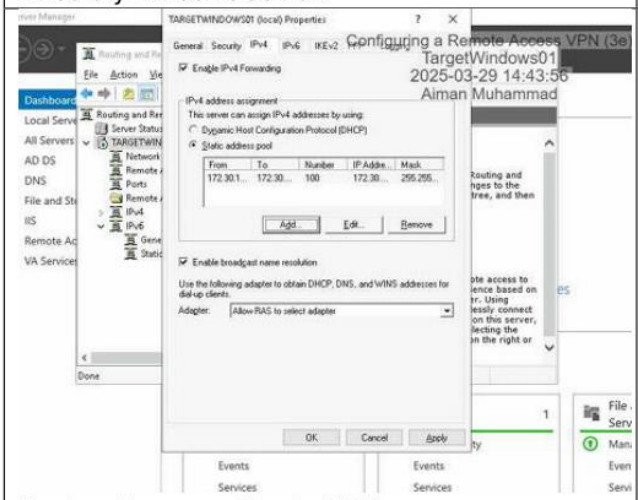
b. Configuring a Remote Access VPN



Remote Access VPN role installed on an Active Directory Windows Server.



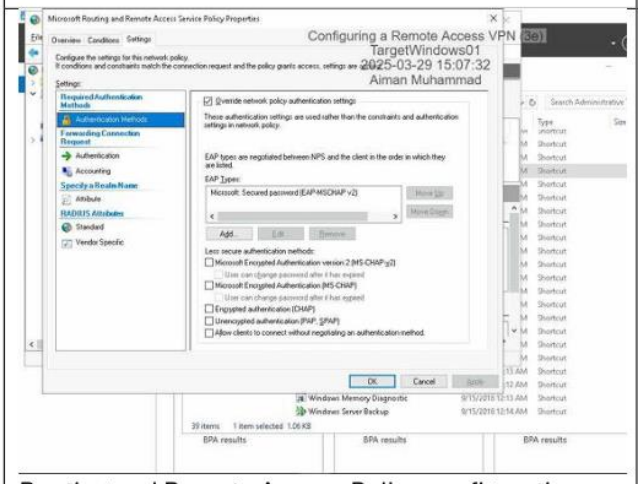
Routing and Remote Access console active on the Windows machine.



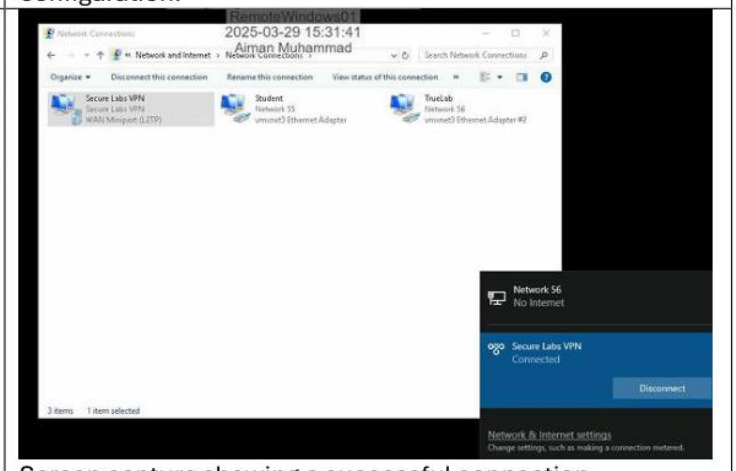
Static address pool for the VPN.



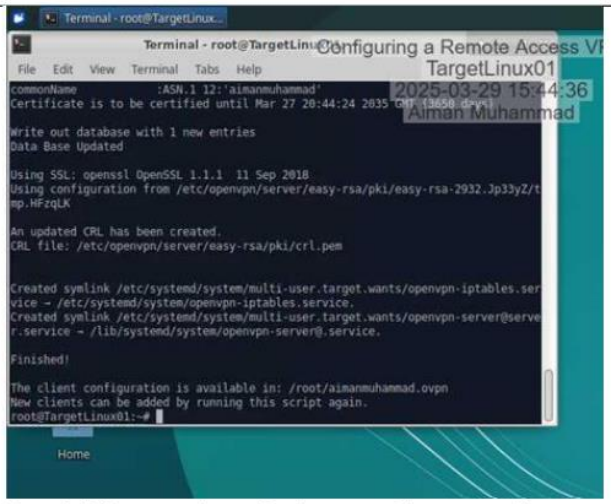
Windows defender Firewall installed for access control configuration.



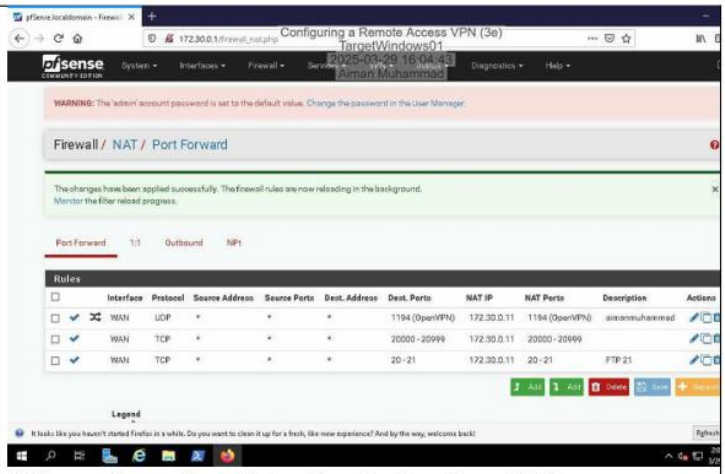
Routing and Remote Access Policy configuration.



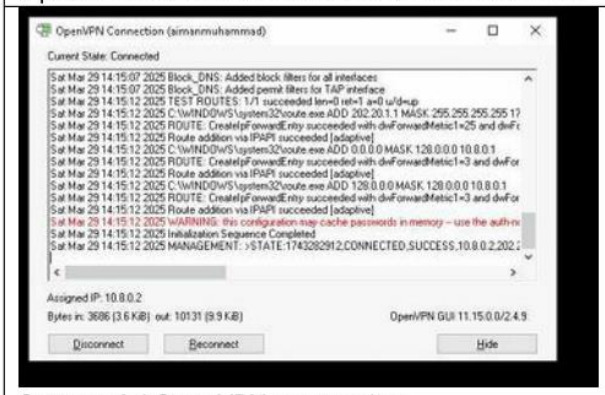
Screen capture showing a successful connection.



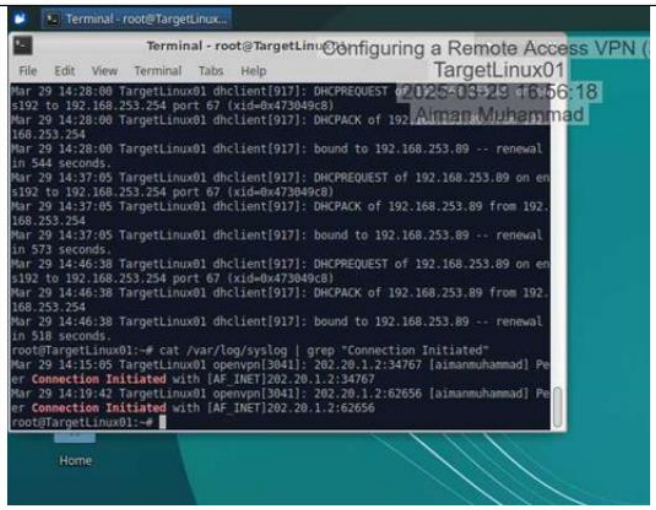
OpenVPN server installation on a Linux machine.



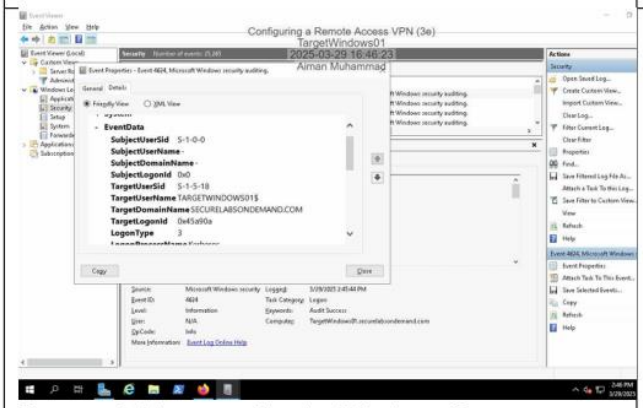
PfSense firewall configuration for the OpenVPN server.



Successful OpenVPN connection.

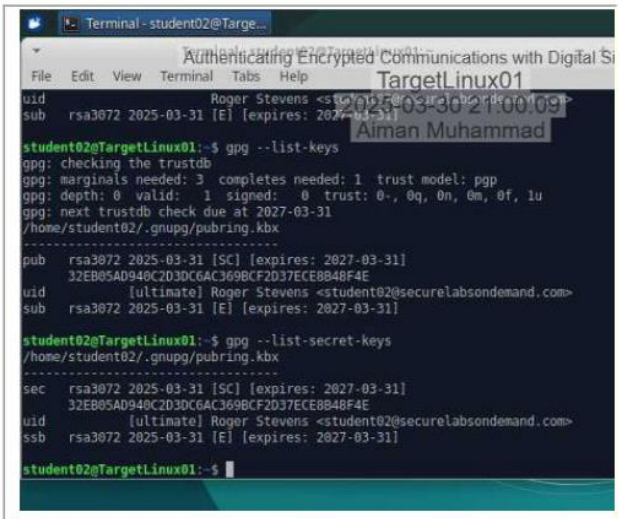


Linux command to view log entries.

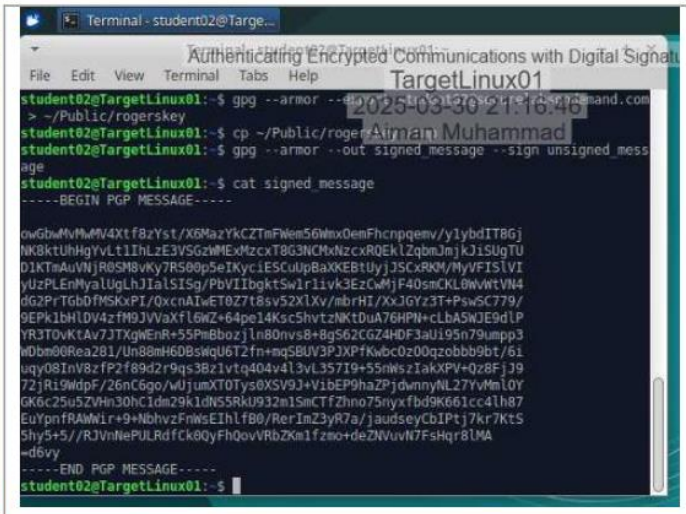


Successful log entry for the Windows Server.

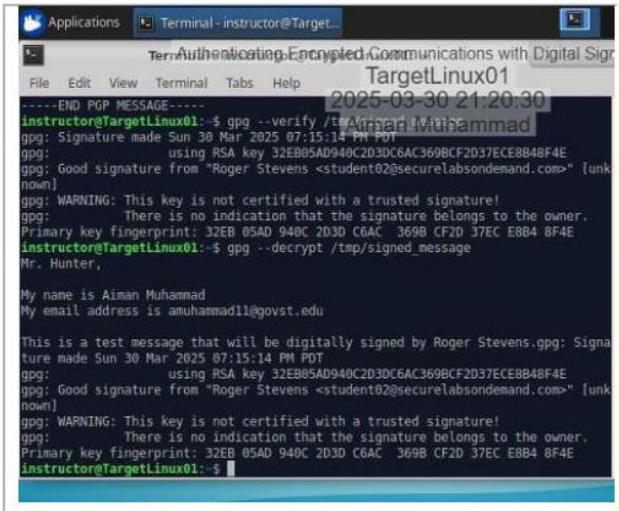
c. Authenticating Encrypted Communication using Digital Signatures



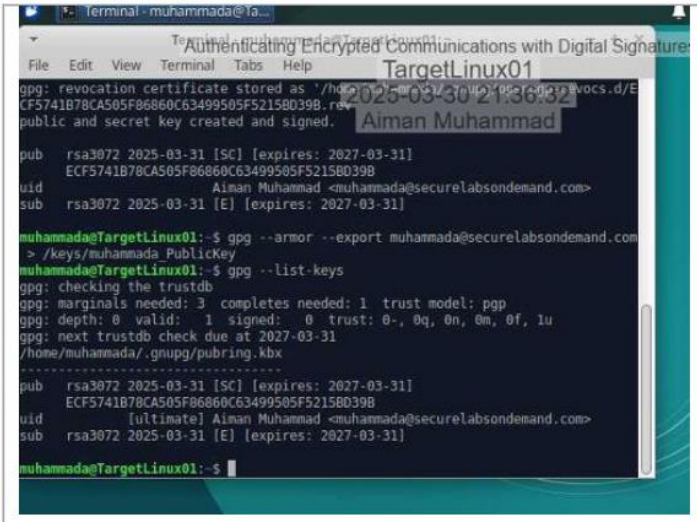
Command that shows the list of Public and Private keys on the linux machine.



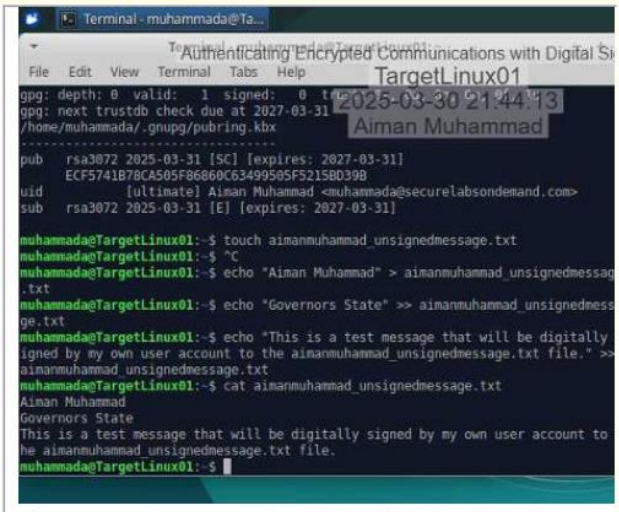
Command showing the encrypted contents of a generated message.



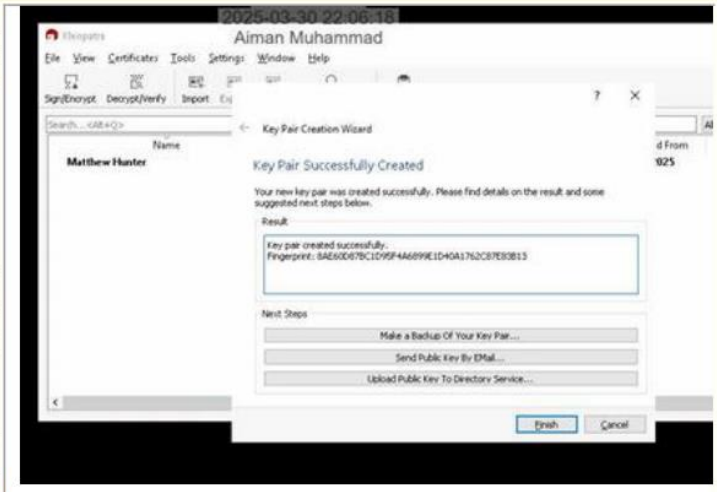
Command that shows the unencrypted contents of the generated message. It currently has no key fingerprint for the digital signature.



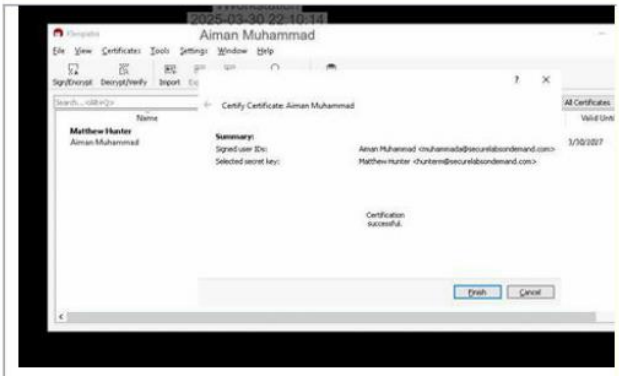
Command to create the key fingerprint for the key pair.



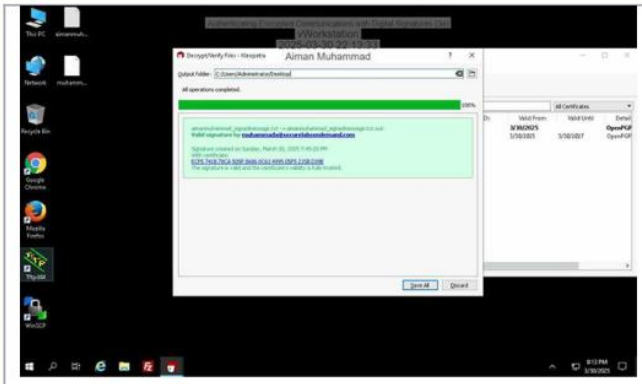
Contents of a created txt file without a digital signature.



Key Pair created using Kleopatra.



Successful certificate created using Kleopatra.



Successful decryption using Kleopatra.



Contents of the decrypted txt file.

Project 4: Wireshark Packet Capture Analysis

Project Overview

Captured and analyzed network traffic using Wireshark to examine how devices communicate across a network. Focused on understanding packet structure, identifying network protocols, and examining the exchange of information between a client and a web server.

Objectives

- Capture and inspect network traffic.
- Analyze TCP, UDP, HTTP, and DNS protocol activity.
- Examine packet headers, source and destination addresses, and port information.
- Understand the communication process between a client and a remote server.

Technologies Used

- Wireshark
- HTTP
- TCP/IP
- Network Packet Analysis

Key Tasks Performed

- Captured network traffic using Wireshark while accessing a web server.
- Applied protocol filters to isolate and examine specific types of network traffic.
- Analyzed HTTP GET requests and corresponding server responses.
- Examined packet details, including source and destination IP addresses, ports, and protocol information.
- Reviewed packet structure to better understand how network communications are established and how information is exchanged.

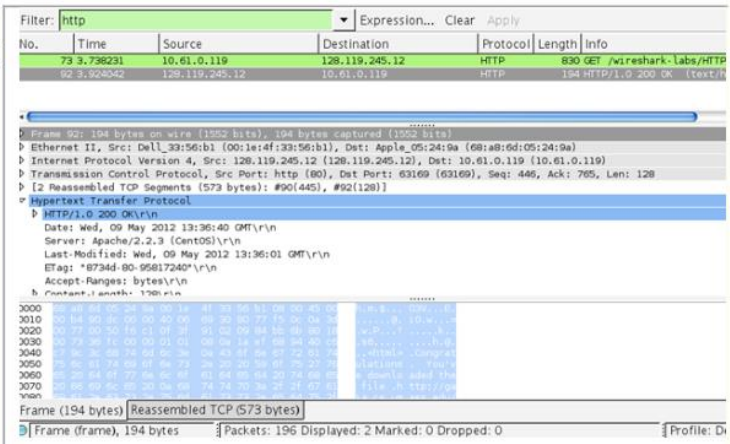
Outcome

Successfully captured and analyzed HTTP traffic between a web browser and a remote web server. Identified the HTTP GET request sent by the client and the corresponding response returned by the server. This exercise strengthened my understanding of packet-level network analysis, protocol behavior, and the role of Wireshark in network troubleshooting.

a. **Wireshark HTTP Packet Capture Analysis**



The HTTP website.



The resulting packet capture from port Wireshark shows in the packet-listing window that two HTTP messages were captured: the GET message from the browser to the gaia.cs.umass.edu web server and the response message from the server to your browser. The packet-contents window shows details of the selected message.

6. Conclusion

This portfolio reflects my ability to apply IT concepts in practical environments, including system administration, networking, and security. I am continuously expanding my skills and seeking opportunities to apply this knowledge in real-world roles.

Appendix A: Example Naming Standards from The University of Wisconsin

Two names will be associated with each domain, DNS and NetBIOS. The DNS name for each domain consists of the host and network names. When these two names are put together they make a unique reference to each domain. The host portion of the name should match the NetBIOS name selected for the domain. The network portion of the DNS name should be a sub domain of a domain name registered to the organization. NetBIOS names will have to be unique within a forest and are most often seen and used by users when authenticating to a particular domain.

Root Domain:

NetBIOS Name - AD

DNS Name - ad.wisc.edu

Child Domains: <depart-code>.ad.wisc.edu

NetBIOS Name - <DEPT>

DNS Name - <DEPT>.ad.wisc.edu

The creation of child active directory domains is not currently allowed under the CADS. If they are determined essential by the operating agency, Active Directory Team will conduct a technical review and validation to authorize any appropriate modifications and naming changes that may be required. That information will then be added to this document and any other appropriate architecture documents.

https://kb.wisc.edu/iam/page.php?id=30600#kb-toc-anchor12

Appendix B: Hypothetical Internet protocol addressing standards

Location	VLAN	Subnet	Department	Summary Address
Headquarters				102.168.0.0/19
	1	192.168.1.0/23	HQ Management	
	8	192.168.8.0/24	HQ Data	
	10	192.168.10.0/24	HQ Wireless Data	
	12	192.168.12.0/24	HQ Voice	
	14	192.168.14.0/24	HQ Wireless Voice	
	16	192.168.16.0/24	Wireless Guest	
	28	192.168.28.0/24	Server Farm A	
	29	192.168.29.0/24	Server Farm B	
	31	192.168.31.0/24	Core Routing	
Branch 1				192.168.64.0/21
	64	192.168.64.0/24	Branch 1 Data	
	65	192.168.65.0/24	Branch 1 Wired Voice	
	66	192.168.66.0/24	Branch 1 Reversed	
	67	192.168.67.0/24	Branch 1 Reversed Security	
	68	192.168.68.0/24	Branch 1 Reversed	
	69	192.168.69.0/24	Branch 1 Wireless Data	
	70	192.168.70.0/24	Branch 1 Wireless Voice	
	71	192.168.71.0/24	Branch 1 loop-backs in 32's	

https://www.cisco.com/c/dam/global/en_ca/solutions/strategy/docs/sbaBN_IPv4addrG.pdf